Enhancing Financial Trading Strategies with Pattern Recognition: A Systematic Literature Review of Methods, Trend and Challenge

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***Abstract*—This study investigates how pattern recognition in financial markets can enhance trading strategies through a systematic literature review (SLR). The research focuses on the most frequently studied financial markets, commonly used pattern recognition methods, identified chart patterns, the integration of trading strategies with pattern recognition, and future challenges. This study is significant because many approaches to pattern recognition analysis are subjective and lack scientific evidence, leading to varied interpretations and uncertainties. The methodology includes formulating research questions, defining inclusion and exclusion criteria, conducting a comprehensive literature search, critically assessing included studies, data extraction and management, and result analysis and interpretation. This SLR covers 46 articles from databases such as Scopus, ACM Digital Library, IEEE Xplore, and Science Direct, published between 2015 and 2024, relevant to the topic. The findings show a dominant use of deep learning technologies, such as Convolutional Neural Networks (CNN), segmentation techniques, and fuzzy logic in recognizing patterns, particularly classic patterns like head and shoulders, triangles, double tops, and others, which contribute to enhancing trading strategies.**

Keywords—pattern recognition, financial chart pattern, trading strategies, convolutional neural networks (CNN), segmentation

I. Introduction

Technical analysis is an approach based on historical data to identify chart patterns and predict future price movements [1]. The appearance of chart patterns serves as a signal to predict future stock price movements [2], thus, the ability to recognize and analyze patterns in charts is crucial for making accurate trading decisions. Technical analysis, also known as "charting," has been a part of financial practices for decades. However, this knowledge has been criticized for its subjective nature and the lack of academic studies supporting it [3]. Traditional technical analysis relies on the visual interpretation of historical price charts, making it susceptible to individual biases. Additionally, these predictive techniques and forecasting strategies often seem unscientific, primarily relying on the experience of seasoned traders [4]. To overcome these limitations, researchers have turned to machine learning algorithms to develop automated and systematic approaches for pattern recognition in financial markets.

This study aims to thoroughly examine the understanding needed to enhance trading strategies in financial markets through chart pattern identification. This SLR is essential as it provides deep insights into current methods, chart patterns, and their application in financial trading. The SLR addresses the following research questions: 1) What types of datasets and which countries are represented in these datasets? 2) What are the methods and publication trends in the field of Chart Pattern Recognition and Classification in Financial Markets? 3) How is the integration of pattern recognition methods with trading strategies used to improve trading performance? 4) Which patterns are identified in financial markets? 5) What are the main challenges faced in applying pattern recognition technology in financial markets?.

II. Methodology

A. Search Strategy and Selection Criteria

This paper focuses on analyzing chart pattern recognition methods and models for improve trading strategy in financial markets.

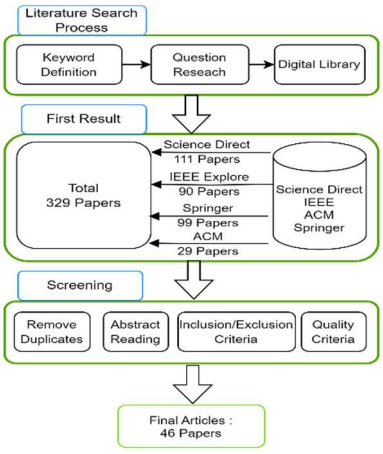


Fig. 1. Literature search and selection

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Publication searches were conducted in several publication databases, including: Springer, ACM Digital Library, IEEE Xplore & Access, and Science Direct. The keywords used in searching are: "financial market" OR "trading strategies" AND ("chart pattern" OR "candlestick pattern" OR "Pattern recognition" OR "stock pattern") AND ("machine learning" OR "Deep Learning")).

The article selection process is carried out in several stages to filter a collection of articles from various databases. The initial search found 329 related publications, furthermore we filtered abstracts, and year of publication between 2015 and 2024 to ensure the suitability of the topic and the latest of the articles then obtained 67 articles. After carrying out a review process, 46 articles were finally used in this paper as shown in Figure 1.

B. Data Extraction Process and Management

The data extraction process is an important step to ensure that the information collected is relevant and reliable. Each selected document will be reviewed and relevant data will be extracted. The extracted data includes publication year, author name, title, chart pattern recognition model used, analyzed dataset, chart pattern identification, identified gaps and future research proposals. We use reference management software or spreadsheets to organize the extracted data.

C. Analisys and Interpretation of Results

Categorize the results based on key themes such as methods for pattern recognition, identifiable patterns, how trading strategies are enhanced through pattern recognition, and research challenges. Compare the study results with existing literature, identifying trends, gaps, or areas that require further research.

D. Published Report

This report consists of the introduction, methodology, results, discussion, and conclusion sections. For clarity, the results are presented in tables and charts. This publication has also been submitted to a journal relevant to the research topic.

III. Result

A. Types and Countries are Represented as Dataset Objects in Financial Market Pattern Recognition

Figure 2 shows the dominance of datasets from the United States financial market in pattern recognition research within financial markets, with 20 datasets utilized [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], followed by China and Hong Kong, each with 6 datasets. Other countries, such as Turkey, Japan, Taiwan, and South Korea, have lower dataset representation, while countries like Saudi Arabia, India, Italy, Germany, the United Kingdom, and Brazil contribute only one dataset or fewer. In addition to country-based datasets, there are also global/multiple and synthetic datasets used to increase data variety. In conclusion, research in this field is highly influenced by data from the U.S. and Asia, with limited representation from other markets.

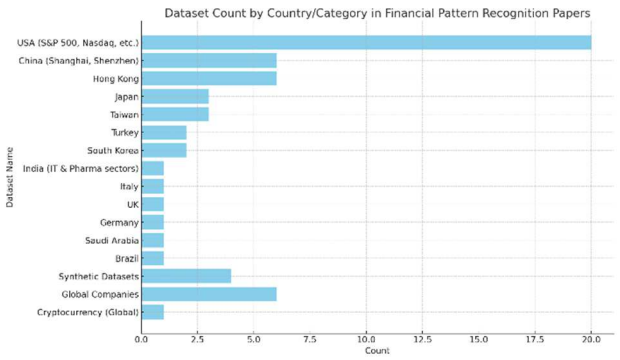


Fig. 2. Number of datasets by country/category in financial pattern recognition research

Based on the types of financial markets shown in Figure 3, the stock market is the most frequently used for research, with 38 studies, followed by the forex market with 5, the commodity market with 2, and cryptocurrency with 1 study. This data indicates a research tendency to use stock market data more extensively than other types of markets.

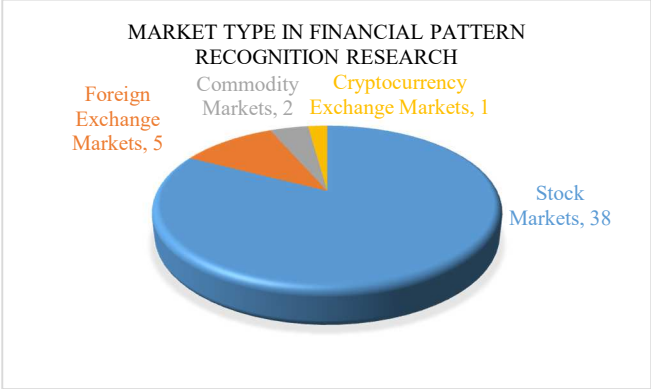


Fig. 3. Market type in financial pattern recognition research

B. Methods and Trends Research on Recognition of Chart Pattern in the Financial Market

This study categorizes pattern recognition methods in financial markets into four main categories. The most frequently used methods fall under the category of Deep Learning and Reinforcement Learning, such as You Only Look Once (YOLO), CNN-Autoencoder, Gramian Angular Field (GAF), Channel and spatial attention convolutional neural network (CS-ACNN), 1D-CNN, Graph Convolutional network (GCN), Double Deep Q-Network (DDQN), Proximal Policy Optimization (PPO), Deep Reinforcement Learning (DRL), Ensemble Learning, Recurrence Plot-based Convolutional Neural Network (RP-CNN), and Deep Candlestick Predictor (DCP) [25], [26], [5], [27], [8], [9], [10], [11], [28], [29], [14], [30], [15], [31], [32], [33], [34] with 17 papers, followed by Rule-Based and Time Series Pattern Matching with 17 papers, using technologies such as Dynamic Time Warping (DTW), Perceptually Important Points (PIP), Rule-Based, Template Matching, Symbolic Aggregate Approximation (SAX) [35], [6], [36], [37], [38], [39], [40], [13], [41], [42], [17], [18], [20], [43], [22], [44], [24] highlighting the dominance of deep learning technologies like Convolutional Neural Networks (CNN) for pattern recognition techniques, as shown in Figure 4. The other categories include Machine Learning and Optimization

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with 7 papers, and Fuzzy Logic and Hybrid Inference Machines with 4 papers.

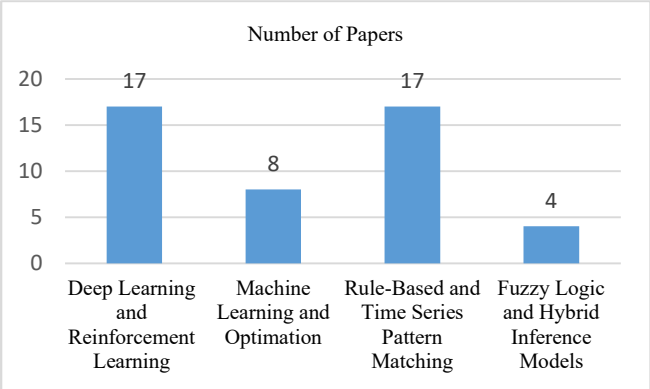


Fig. 4. Distribution graphic pattern recognition method

C. *Integration of Pattern Recognition Methods with Trading Strategies*

The integration of pattern recognition methods with trading strategies can enhance trading performance in various ways, as shown in Table 1. The combination of deep learning, machine learning, rule-based pattern matching, and fuzzy logic methods in pattern recognition within financial markets provides robust support for trading decision strategies by improving prediction accuracy and adaptability to market changes. Deep learning methods, such as CNN and reinforcement learning (YOLO, DDQN, DRL), enable complex candlestick pattern analysis, helping traders quickly identify buy or sell signals in real-time and enhancing prediction stability through ensemble approaches. Traditional machine learning techniques, such as SVM, Random Forest, and optimization based on Genetic Algorithm (GA), offer capabilities for analyzing recurring patterns and optimizing trading parameters to minimize risks. Rule-based and pattern matching methods (e.g., DTW, PIP) improve accuracy by detecting classic patterns like head and shoulders, providing additional guidance for rule-based technical strategies. Additionally, fuzzy logic models like ANFIS allow flexible interpretation of ambiguous price patterns, enhancing the system's ability to make adaptive decisions based on dynamic market conditions. Overall, the integration of pattern recognition methods with trading strategies not only accelerates and refines trading decisions but also optimizes profitability and reduces risks by providing more timely, data-driven buy or sell signals.

TABLE I. METHODS AND APPROACHES IN INTEGRATING PATTERN RECOGNITION WITH TRADING STRATEGIES

No	Technology Category	Description	Citation
11	Deep Learning and Reinforcement Learning	Various algorithms, including CNN, GCN, and reinforcement learning, are used for candlestick pattern detection, trading strategy optimization, and prediction stability through ensemble learning.	[25],[26],[5], [27],[8],[9],[10], [11],[28],[29], [14],[30],[15], [31],[32],[33], [34]
22	Machine Learning and Optimization	Traditional machine learning algorithms and GA-based optimization are used for detecting stock chart patterns and technical indicators, as well as grid optimization in pattern detection.	[4],[7],[45], [46],[16],[19], [21],[23]
33	Rule-Based and Time Series Pattern Matching	The use of template-based rules, time series segmentation, and financial data processing techniques enables the detection of classic chart patterns and time-based patterns.	[35],[6],[36], [37],[38],[39], [40],[13],[41], [42],[17],[18], [20],[43],[22], [44],[24]
44	Fuzzy Logic and Hybrid Inference Models	Fuzzy-based and hybrid inference models are used for detecting complex chart patterns, converting images from time series data, and identifying candlestick patterns.	[47],[43],[48], [49]

D. *Types of Identified Chart Patterns*

Chart patterns frequently used as examples in pattern recognition include classic patterns such as *Head and Shoulders*[4], [6], [8], [36], [39], [40], [13], [41], [43], [21], [32], [24] and *Double Top/Bottom* [35], [6], [36], [39], [40], [13], [21], [32] which are utilized to detect trend reversals and often serve as references in technical analysis [4],[35],[32]. Two papers cover 53 patterns categorized according to Thomas Bulkowski [50], as identified in [48],[20]. Patterns like Bull Flag/Bear Flag are commonly used to detect trend continuation or consolidation [6],[16],[22],[33]. In addition to classic patterns, some studies document new or non-classic patterns, such as rising-support patterns, which offer a unique approach to identifying market trends not explained by traditional patterns [7], [28], [37], [46], [29], [30], [31], [34]. There are also papers employing automated or general trend-based approaches without specifying particular patterns, focusing on model adaptation to market conditions without referencing specific classic patterns [26], [9], [38], [47], [12], [14], [15], [42], [19]. These categories illustrate the diversity of approaches in applying chart patterns to support trading decisions in financial markets.

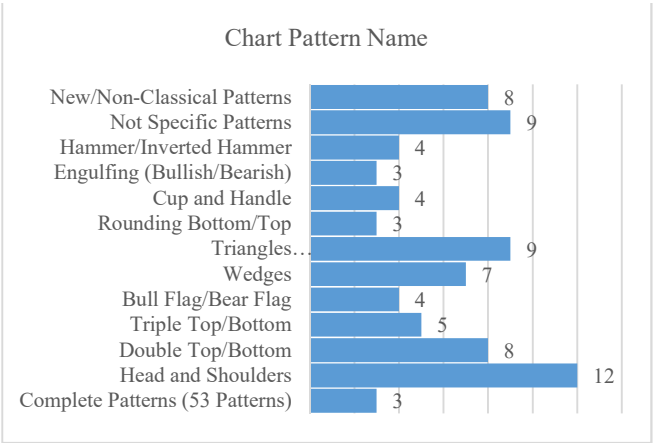


Fig. 5. Types of identified chart patterns

E. *The Main Challenges in Implementing Pattern Recognition Technology in Financial Marketsl*

Pattern recognition model challenges in financial markets, based on findings from previous studies, highlight five main areas for future research. First, expanding datasets

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and integrating external data, such as technical indicators, news sentiment, and fundamental factors, can enhance model accuracy and scope [25], [5], [7], [46], [47], [29], [30], [15], [23]. Second, optimizing models to be more robust and interpretable, including automated hyperparameter tuning, can improve accessibility as well as resilience to market volatility and adversarial attacks [35], [6], [9], [10], [45], [29], [31]. Third, the application of ensemble techniques and reinforcement learning makes models more adaptive, ensuring stable performance under extreme market conditions [8], [9], [28], [37], [46], [13], [31]. Fourth, developing advanced time series segmentation and pattern matching techniques enables better handling of complex, real-time, and high-frequency data [8], [49], [38], [39], [48], [42], [44], [32]. Lastly, improving computational efficiency and expanding model applications to other markets, such as Forex and commodities, ensures model effectiveness across diverse environments [27], [11], [28], [13], [14], [41], [20], [22], [33].

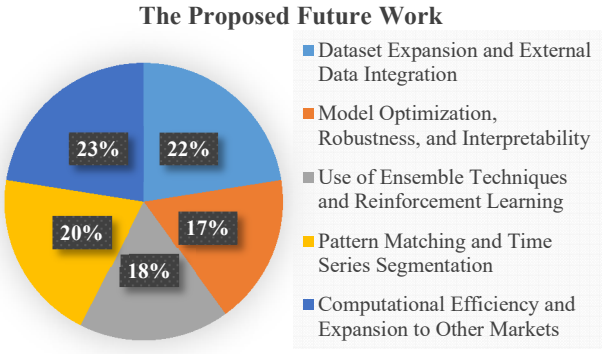


Fig. 6. The proposed future work

IV. Discussion

A. Interpretation of Finding

The findings of this study reveal that pattern recognition in financial markets can enhance trading strategies through various artificial intelligence techniques. Deep learning technologies, such as CNN and its variants [25], [27], [8], [9], [10], [11], [36], [29], [30], [15], [19], [31] and reinforcement learning [26],[14],[9],[37] enable complex pattern recognition and produce more stable predictions, while classical machine learning algorithms assist in analyzing recurring patterns and optimizing parameters [4], [7], [45], [46], [16], [19], [21], [23]. Additionally, rule-based approaches and pattern matching using methods such as Dynamic Time Warping (DTW) add accuracy in detecting classic patterns [38],[12],[42]. More complex patterns can also be identified through fuzzy logic models [17], [43], allowing for a more flexible interpretation of ambiguous price data. The integration of these methods supports adaptive, data-driven trading strategies, providing more accurate signals and assisting traders in decision-making within financial markets.

B. Comparison with the existing literature

This review adds to the existing literature by analyzing the application of newer machine learning and deep learning methods in pattern recognition within financial markets [36],[29],[9],[27]. The study demonstrates that, while previous research has primarily focused on classic patterns such as head and shoulders, triangle, and double top/bottom [4], [6], [8], [36], [39], [40], other methods allow for more varied and flexible pattern recognition [25], [26], [7], [37]. Compared to classic rule-based approaches, the integration of machine learning-based methods in more recent literature tends to reduce subjectivity and enhance accuracy in identifying patterns by accounting for more complex market dynamics.

C. Strengths and Limitations of the Review

This study successfully summarizes various approaches in pattern recognition, providing a comprehensive understanding of recent developments, their applications in trading strategies within financial markets, and opportunities for ongoing innovation, especially in identifying chart patterns beyond classic ones [25], [26], [7], [37], [49], [28], [45], [38], [47], [12], [14]. However, the review has limitations, including a high representation of U.S. market-based datasets and stock market data, which may not fully reflect the dynamics of other markets and types. Additionally, challenges such as adapting methods to different markets, optimizing model parameters, and difficulties in result interpretation still require further exploration. The reliance on available literature can constrain the adequacy of conclusions, and the continually evolving literature may necessitate future updates to this review.

V. Conclusions

Pattern recognition in financial markets plays a crucial role in enhancing trading strategies. Methods such as deep learning, machine learning, and rule-based approaches have been used to identify classic patterns, like "head and shoulders," "triangle," and "double top/bottom." These patterns have proven effective in supporting more precise trading decisions. Furthermore, research has identified new or non-classic patterns that are more flexible and can adapt to the ever-changing dynamics and complexity of the markets. This demonstrates the potential of integrating these new methods to support enhanced stability, accuracy, and adaptability in pattern recognition-based trading systems.

Future research should focus on several aspects, particularly in optimizing datasets, exploring segmentation techniques, and integrating various data sources, such as sentiment analysis and fundamental indicators. Exploring ensemble or hybrid models, improving hyperparameter tuning, and incorporating attention mechanisms are also expected to enhance pattern detection accuracy. Additionally, expanding research to include different markets, such as cryptocurrencies or commodities, will strengthen models' capabilities to handle volatility and diverse market conditions, resulting in a more robust and sustainable pattern recognition system, thereby supporting the development of more optimal and adaptive trading strategies across various market conditions.

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